A Structural Local Identifiability Obstruction for the 31-Coefficient v18 Spectral Model

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Abstract

We consider the eleven-nonzero-observable map of the v18 spectral model with twenty-nine stored scalar coefficients and two historically optimized parameters. At every differentiability point, rank–nullity implies at least twenty first-order null directions. For the exact-decimal frozen coefficients, $\epsilon = 5$, and $\alpha \in [5.172, 5.173]$, we additionally prove that the actual clipped 31-parameter map is C^1 near the frozen vector, its derivative has rank exactly eleven and nullity exactly twenty, and its local equal-output fiber is a twenty-dimensional C^1 embedded submanifold. The proof uses a selected minor of the full Fréchet derivative, a kernel-checked trigonometric determinant enclosure, and the implicit-function theorem. All mathematical claims are formalized in Lean 4 with Mathlib v4.23.0. The historical finite-difference SVD remains a separate numerical diagnostic. This is an audit of parameter identifiability, not an independent physical prediction of particle masses.

1 Model and hypotheses

Let $U \subseteq \mathbb{R}^{31}$ be an open neighborhood of a reference coefficient vector θ_0, and let

$$F : U \longrightarrow \mathbb{R}^{11}, \qquad F(\theta) = \big(f_1(\theta), \ldots, f_{11}(\theta)\big)^{\mathsf{T}} \tag{1}$$

be the spectral forward map for the eleven nonzero observables. The assumption that F is differentiable at θ_0 is explicit: the implementation includes piecewise routing, clipping, and floors, so differentiability should not be asserted at a branch transition. Within a fixed smooth branch, the calculation below applies directly. The rank–nullity conclusion also applies to any chosen linear sensitivity matrix of the indicated shape, independently of how its entries were estimated.

The audited parameter allocation is

Parameter section	Number of scalar parameters
Light sector	9
Middle sector	7
Heavy sector	13
Historically live suppression parameters	2
Total	31

The v18 source specifies twelve targets, including one artificial zero-mass target. The full-coefficient audit excludes that zero target and uses the index set

$$\mathcal{N} = \{1, 2, 3, 12, 19, 53, 73, 91, 95, 96, 123\}, \qquad |\mathcal{N}| = 11. \tag{2}$$

The corresponding eleven outputs are evaluated without refitting the stored coefficients or reassigning particle labels.

For the concrete certificate, θ_0 is the vector `V18Forward.frozen alpha 5`. Its twenty-nine stored coefficients are the published finite decimals, interpreted as exact real numbers; $\alpha \in [5.172, 5.173]$. The Lean function `V18Forward.observed` retains all thirty-one input coordinates, including both middle-sector frequencies, both suppression parameters, all heavy powers, and the W/Z frequency and phase. It includes the original floors, exponential and power clipping, and routing. No equivalence with arbitrary NumPy floating-point executions is claimed.

2 Exact structural theorem

Theorem 2.1 (Minimum first-order nullity)**.** *Suppose the map F in* (1) *is differentiable at $\theta_0 \in U$. Its sensitivity matrix*

$$J = DF(\theta_0) = \left(\frac{\partial f_i}{\partial \theta_j}(\theta_0)\right)_{\substack{1\le i\le 11\\ 1\le j\le 31}} \in \mathbb{R}^{11\times 31} \tag{3}$$

satisfies

$$\boxed{\operatorname{rank} J \le 11}, \qquad \boxed{\dim\ker J = 31 - \operatorname{rank} J \ge 20}. \tag{4}$$

In particular, there are at least twenty linearly independent directions in parameter space along which all eleven predictions have vanishing first directional derivative.

Proof. The derivative $DF(\theta_0)$ is a linear map

$$J : \mathbb{R}^{31} \longrightarrow \mathbb{R}^{11}. \tag{5}$$

Because the dimension of its codomain is eleven,

$$\operatorname{rank} J \le \min\{31, 11\} = 11. \tag{6}$$

The rank–nullity theorem gives

$$\dim\ker J + \operatorname{rank} J = \dim \mathbb{R}^{31} = 31, \tag{7}$$

and hence

$$\dim\ker J = 31 - \operatorname{rank} J \ge 31 - 11 = 20. \tag{8}$$

Choose linearly independent vectors $v_1, \dots, v_d$ forming a basis of $\ker J$, where $d \ge 20$. For any $v \in \ker J$, we have $Jv = 0$. Differentiability entails

$$F(\theta_0 + tv) = F(\theta_0) + tJv + o(|t|) = F(\theta_0) + o(|t|) \quad (t \to 0). \tag{9}$$

Thus the first derivative of every component of F vanishes along each v_i, proving the last assertion. □

Remark 2.2 (Meaning of "unidentifiable direction")**.** *Equation* (9) *is a* first-order *statement; it neither says that a particular parameter is independently dispensable nor that $F(\theta_0 + tv) = F(\theta_0)$ holds for nonzero t. The null vectors are generally combinations of the stored and live coefficients. Furthermore, if the physically admissible parameter set has active bounds, not every signed null perturbation need remain feasible. The theorem concerns the ambient linearized map.*

Corollary 2.3 (Exact nullity under exact full row rank)**.** *If, in addition,* $\operatorname{rank} DF(\theta_0) = 11$ *holds* exactly*, then*

$$\boxed{\dim\ker DF(\theta_0) = 20}. \tag{10}$$

Proof. Insert $\operatorname{rank} J = 11$ into rank–nullity. □

3 Reproducing the numerical sensitivity audit

The full-coefficient script loads the unchanged v18 implementation, checks the original predictions against an independent explicit branch-dispatch evaluator, and applies finite differences about the historical parameter vector θ_0. For a coordinate vector e_j, it usually estimates

$$J_{ij} \approx \frac{f_i(\theta_0 + h_j e_j) - f_i(\theta_0 - h_j e_j)}{2h_j}, \qquad h_j = 10^{-5} \max\{1, |\theta_{0,j}|\}. \tag{11}$$

The historically live parameter ϵ_{cutoff} is at its upper bound 5, so that column is computed using a backward difference. Other active bounds, when relevant, also trigger one-sided differences. These finite differences are diagnostics and are not an exact derivative certificate.

Let $y_i > 0$ denote the eleven observed nonzero target masses and set $s_j = \max\{1, |\theta_{0,j}|\}$. The reported scaled matrix is

$$\widetilde{J} = D_y^{-1} J D_s, \qquad D_y = \operatorname{diag}(y_1, \ldots, y_{11}), \quad D_s = \operatorname{diag}(s_1, \ldots, s_{31}). \tag{12}$$

Both diagonal matrices are invertible, so exact linear-algebraic rank is unchanged:

$$\operatorname{rank} \widetilde{J} = \operatorname{rank} J. \tag{13}$$

However, a numerical rank estimate can depend on scaling, finite differences, and the threshold used in the SVD.

Writing the eleven singular values as $\sigma_1 \geq \cdots \geq \sigma_{11} \geq 0$, the script defines

$$r_{\text{num}} = \#\{i : \sigma_i > 10^{-8} \sigma_1\}. \tag{14}$$

The executable audit reports

$$\boxed{r_{\text{num}} = 11}, \qquad \boxed{31 - r_{\text{num}} = 20}. \tag{15}$$

This is *numerical* full row rank. By contrast, the lower bound $\dim \ker J \geq 20$ in Theorem 2.1 is an exact consequence of dimensions, even if the true rank were below eleven.

4 Exact selected-minor certificate

The proof selects the following zero-based coordinates of the complete 31-dimensional parameter vector, ordered by light, middle, and heavy sector:

$$(0, 3, 4;\ 14, 15, 9, 12, 13;\ 20, 24, 28). \tag{16}$$

These are, respectively, (m_0, θ, γ), middle (c, d, A, ϕ, h), and heavy (first nonlinear boson scale, W/Z-break scale, Higgs offset). This selection differs from the earlier finite-difference diagnostic's columns. It changes neither the forward map nor its active parameters; it selects directions only *after* differentiating the complete map.

Strict inequalities at the frozen vector fix every relevant clipping branch on an open neighborhood. All three light outputs and all five middle outputs exceed their floors strictly. At the three heavy indices, the inner light prediction is strictly below its floor, so that inner term is locally the constant 10^{-12}; each heavy outer prediction is strictly above its floor. The relevant exponential and power clips are also strictly inside, or strictly on the verified saturated side of, their thresholds. Continuity propagates these inequalities locally.

Consequently, the selected restriction of the actual Fréchet derivative is

$$J_{\text{minor}} = \begin{pmatrix} L & 0 & 0 \\ 0 & M & 0 \\ 0 & 0 & H \end{pmatrix}, \qquad \det J_{\text{minor}} = \det L \, \det M \, \det H. \tag{17}$$

All 121 entries, including all six off-diagonal blocks, are identified in Lean. In particular, the heavy-to-light coupling vanishes because of the verified inner floor, not because light coefficients have been held fixed in the full derivative.

4.1 Light sector

The selected light block is $L = \operatorname{diag}(D_1, D_2, D_3)L_0$, where

$$L_0 = \begin{pmatrix} 1/2 & 1 & 1 \\ 2/3 & 4 & 8 \\ 3/4 & 9 & 27 \end{pmatrix}, \qquad D_n = \frac{\eta_n}{1 + \exp(\alpha(n-5))}, \qquad (\eta_1, \eta_2, \eta_3) = (1, 1, 0.0048). \tag{18}$$

Exact rational expansion gives

$$\det L_0 = 9, \qquad \det L = 9 D_1 D_2 D_3 > 0. \tag{19}$$

4.2 Middle sector

The unfloored formula is

$$f(n) = c + dn^2 + h \sin(tn) + A \sin(sn + \phi). \tag{20}$$

For $n \in \{12, 19, 53, 73, 91\}$, the selected derivative rows are

$$M_n = (1, n^2, \sin(sn + \phi), A\cos(sn + \phi), \sin(tn)), \tag{21}$$

with $(A, s, t, \phi) = (2241.1, 5.0632, 9.4062, 1.6469)$ at the frozen vector. The frequencies remain active in the complete derivative. The kernel-checked result for this exact matrix is

$$\boxed{18{,}000{,}000 < \det M < 24{,}000{,}000}. \tag{22}$$

The proof uses rational bounds on π, exact period reduction, Mathlib sine and cosine Taylor remainder estimates, and rational interval propagation through an exact 60-term determinant expansion. External Python certificates are regression checks and supply no Lean premises.

4.3 Heavy sector

Let $a_n = (n/100)^{-1.189}$. The selected heavy block is

$$H = \begin{pmatrix} a_{95} & \sin(-115.395) & 0 \\ a_{96} & \sin(-116.659) & 0 \\ a_{123} & 0 & 1 \end{pmatrix}. \tag{23}$$

Here the first column differentiates the first nonlinear boson scale, the second the W/Z-break amplitude, and the third the Higgs offset. The full derivative also retains the power, frequency, and

phase derivatives. Period reduction with certified rational bounds on π proves $\sin(-115.395) < 0 < \sin(-116.659)$. Since $a_{95}, a_{96} > 0$,

$$\det H = a_{95} \sin(-116.659) - a_{96} \sin(-115.395) > 0. \tag{24}$$

Thus the actual selected derivative restriction is nonsingular. Its rank is eleven and cannot exceed the rank of the full derivative. The latter has eleven output coordinates, so

$$\boxed{\operatorname{rank} DF(\theta_0) = 11}, \qquad \boxed{\dim \ker DF(\theta_0) = 20}. \tag{25}$$

5 Local level-set geometry

Proposition 5.1 (Conditional nonlinear local level set)**.** *Assume F is continuously differentiable on an open neighborhood of θ_0, and assume the* exact *rank condition* $\operatorname{rank} DF(\theta_0) = 11$*. Then, in a sufficiently small neighborhood of θ_0, the level set*

$$\mathcal{M} = \{\theta : F(\theta) = F(\theta_0)\} \tag{26}$$

is a C^1 submanifold of dimension twenty.

Proof. Exact row rank eleven means that $DF(\theta_0)$ is surjective. The submersion (or implicit-function) theorem then provides local coordinates in which F is the projection from $\mathbb{R}^{31}$ onto the first eleven coordinates, up to smooth changes of coordinates. The fiber over $F(\theta_0)$ is locally described by fixing those eleven coordinates, leaving $31 - 11 = 20$ free coordinates. □

Corollary 5.2 (The concrete frozen fiber)**.** *For the exact frozen vector with $\alpha \in [5.172, 5.173]$ and $\epsilon = 5$, all hypotheses of Proposition 5.1 hold. More precisely, writing $K = \ker DF(\theta_0)$, there is an ambient partial homeomorphism Ψ between open neighborhoods in $\mathbb{R}^{31}$ and $\mathbb{R}^{11} \times K$ such that*

$$\Psi(\theta_0) = (F(\theta_0), 0), \qquad \pi_1 \Psi(\theta) = F(\theta), \qquad \dim K = 20. \tag{27}$$

Both Ψ and Ψ^{-1} are C^1 on their open domains. The chart maps the entire local fiber exactly onto

$$\{(y, k) \in \operatorname{target}(\Psi) : y = F(\theta_0)\}.$$

Every ambient neighborhood of θ_0 contains a distinct parameter vector with exactly the same eleven predictions.

Proof. The verified strict branches give C^1 regularity on an open neighborhood. The certified minor proves surjectivity of the full Fréchet derivative. Mathlib's implicit-function chart is restricted to open sets where the chart and its inverse are C^1; the inverse regularity follows from its inverse-function theorem. The first chart coordinate is the actual observation map, and the second takes values in its twenty-dimensional derivative kernel. Varying a nonzero kernel coordinate slightly proves the final assertion. These steps are exported by `V18Forward.f09_completion` and `V18Forward.observed_locally_nonidentifiable`. □

Remark 5.3 (Active bounds)**.** *The historical live parameter ϵ_{cutoff} sits at an optimizer upper bound. Proposition 5.1 is therefore stated for a smooth extension to an* open *ambient domain; it does not automatically imply that every direction in the full twenty-dimensional fiber respects the fitted parameter constraints. Since the certified minor in Section 4 uses eleven* stored *parameters, it also permits a restricted implicit-function analysis with the two live parameters fixed. In that case the remaining twenty-nine stored parameters would support a local $29 - 11 = 18$-dimensional equal-prediction level set, without moving either live parameter across its bound.*

6 Scope and scientific conclusion

The theorem makes no claim that exactly twenty individually named coefficients are redundant, that the measured masses identify the remaining eleven coefficients, or that a fitted spectral formula follows from a specified eigenvalue operator. It does not determine measurement uncertainties or Bayesian posterior widths. In particular, it does not supply an independent mapping from eigenmodes to particle assignments, a confinement operator with its domain and boundary conditions, or a physical normalization converting eigenvalues into measured mass units.

The logical status of the main statements is:

Statement	Status
$\operatorname{rank} J \leq 11$ for an 11-by-31 Jacobian	Exact theorem
$\dim \ker J \geq 20$	Exact theorem
At least twenty zero first-order sensitivities	Exact when differentiable
The frozen audit's reported numerical rank eleven	Numerical diagnostic
Frozen rank eleven and nullity twenty	Kernel-checked exact result
Exactly twenty-dimensional smooth local fibers	Kernel-checked at the frozen vectors
Independent spectral prediction of particle masses	Not established

Conclusion. For the unrestricted 31-coefficient parameterization and eleven nonzero observables, the first-order structural obstruction is

$$\boxed{J \in \mathbb{R}^{11\times 31} \quad \Longrightarrow \quad \operatorname{rank} J \leq 11 \quad \Longrightarrow \quad \dim \ker J \geq 20.} \tag{28}$$

The concrete frozen result additionally establishes a twenty-dimensional nonlinear local fiber. It does not classify distant solutions, determine the global topology of a fiber, or make a physically constrained parameter box into an open manifold. None of these mathematical results constitutes an independently validated physical spectrum. The corresponding v18 research classification remains `EMPIRICAL_FIT_ONLY`.

Sources and reproducibility

1. v18 audit and scientific-status discussion:
   `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/SM_V18_SPECTRAL_BRIDGE_AUDIT.md`
2. Full-coefficient finite-difference/SVD audit:
   `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/SM_Spectrum/audit_v18_identifiability.py`
3. Original v18 source defining the coefficient blocks and predictions:
   `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/SM_Spectrum/_terms_perfection_v18.py`
4. Foundational Stage 2 spectral-bridge obstruction report:
   `https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/FOUNDATIONAL_STAGE2_2026-10-14.md`

The repository's reproduction command is

```
python SM_Spectrum/audit_v18_identifiability.py \
  --output audit/v18_identifiability.json
```

It records coefficient and CSV file hashes, package versions, the 11-by-31 scaled sensitivity, its SVD, and the declared numerical-rank threshold. The historical numerical result is reported separately from the Lean certificate above. No new executed holdout test is claimed.

The main formalization modules in `lean/MathProofs/` are:

- `V18MiddleDeterminant`: the exact trigonometric enclosure.
- `V18Regularity` and `V18Frechet`: local branches and the full derivative.
- `V18SelectedJacobian` and `V18Rank`: the actual minor, rank and nullity.
- `LocalLevelSet` and `V18Identifiability`: the C^1 chart and completion theorem.

The aggregate command is

```
lake build MathProofs.V18Identifiability
```

The complete repository is checked with `lake build`.